

8-Puzzle using IDS

IDS(start, goal, maxDepth)

for $x=0$ to maxDepth

call DLS(state, goal, depth)

if solution found:

return it

return "No solution found".

DLS(state, goal, depth)

if state == goal:

return "Success".

if depth == 0:

return "Failure".

Generate all valid child state

for each child:

call DLS(child, goal, depth+1)

if success:

return "Success"

return "Failure"

1	2	3
	4	6
7	5	8

①

1	2	3
4		6
7	5	8

①

1	2	3
7	4	6
	5	8

	2	3
1	4	6
7	5	8

1	2	3
4	5	6
7		8

1	2	3
4	6	
7	5	8

1		3
4	2	6
7	5	8

①

1	2	3
4	5	6
7	8	

1	2	3
4	5	6
	7	8

goal state

8

path cost = 3

Output

Enter initial state:

1	2	3
0	4	6
7	5	8

Enter goal state:

1	2	3
4	5	6
7	8	

Solution found!

Depth : 3.

Path Cost : 3

Path:

1	2	3		1	2	3		1	2	3
0	4	6	→	4	0	6	→	4	5	6
7	5	8		7	5	8		7	0	8

→	1	2	3
	4	5	6
	7	8	0



8-Puzzle using A*

A* (start, goal):

OPEN = {start}

while OPEN is not empty:

current = node in OPEN with lowest $f(n)$

if current == goal:

return path to goal

remove current from OPEN

for each neighbor of current:

calculate new path cost $g(\text{neighbor})$

if this path is better:

update parent of neighbor

update $g(\text{neighbor})$

$f(\text{neighbor}) = g(\text{neighbor}) + h(\text{neighbor})$

add neighbor to OPEN if necessary

return Failure.

initial state

2	8	3
1	6	4
	7	5

$g(n) = 0$

$h(n) = 0$

goal state

1	2	3
8		4
7	6	5



2	8	3	2	8	3
1	6	4		6	4
7		5	1	7	5

$g(n) = 1$

$h(n) = 4$
 $f(n) = 5$

$h(n) = 5$
 $f(n) = 6$

2	8	3	2	8	3
1		4	1	6	4
7	6	5	7	5	

$g(n) = 2$

$h(n) = 3$

$h(n) = 5$

$f(n) = 5$

$f(n) = 7$

2		3	2	8	3
1	8	4		1	4
7	6	5	7	6	5

$g(n) = 3$

$h(n) = 3$

$h(n) = 3$

$f(n) = 6$

2		3
1	8	4
7	6	5

	2	3			2	3	
1	8	4			1	8	4
7	6	5			7	6	5

$$h(n) = 2$$

$$f(n) = 6$$

$$h(n) = 4$$

$$f(n) = 8$$

1	2	3
	8	4
7	6	5

$$h(n) = 1$$

$$f(n) = 6$$

1	2	3
8		4
7	6	5

$$h(n) = 0$$

$$f(n) = 6$$

1	2	3
7	8	4
	6	5

$$h(n) = 2$$

$$f(n) = 8$$

Goal state

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Output :-

Enter the configuration for initial state

2	8	3
1	6	4
0	7	5

Enter the configuration for goal state

1	2	3
8	0	4
7	6	5

Optimal Solution Found in 6 moves!

step 0

2	8	3
1	6	4
-	7	5

$$g(n) = 0$$

$$h(n) = 6$$

$$f(n) = 6$$

step 1

2	8	3
1	6	4
7	-	5

$$g(n) = 1$$

$$h(n) = 5$$

$$f(n) = 6$$

step 2

2	8	3
1	-	4
7	6	5

$$g(n) = 2$$

$$h(n) = 4$$

$$f(n) = 6$$

step 3

2	-	3
1	8	4
7	6	5

$$g(n) = 3$$

$$h(n) = 3$$

$$f(n) = 6$$

step 4

-	2	3
1	8	4
7	6	5

$$g(n) = 4$$

$$h(n) = 2$$

$$f(n) = 6$$

step 5.

1 2 3

- 8 4

7 6 5

$$g(n) = 5$$

$$h(n) = 1$$

$$f(n) = 6$$

step 6

1 2 3

8 - 4

7 6 5

$$g(n) = 6$$

$$h(n) = 0$$

$$f(n) = 6$$